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Experimental Research on Multi-Scale Micro-Fractures Channeling Model

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Abstract

The sustained annular pressure (SCP) in complex oil and gas wells, such as high temperature high pressure gas wells and gas storage wells, directly affects the safe production and lifecycle. The failure of wellbore integrity in SCP well is hard to observe, and the multi-scale channeling paths are difficult to be measured through logging. Therefore, it is necessary to establish a multi-scale channeling experimental platform. By studying controllable micro-fracture prefabrication technology, batch processing of cements with specific micro-fracture height is achieved. The relationship between channeling flow rate and pressure loss in cement stones with different micro-fracture height is determined, and the equivalent relationship between micro-fracture height and comprehensive equivalent permeability is inverted. A multi-scale micro-fracture channeling model is established based on two main channeling types: macroscopic gas flow in micro-fracture channels and gas seepage in weakly cemented channels. The model is refined using multi-scale channeling experimental data. The research results enable quantitative analysis of annular pressure test data and prediction of annular pressure, providing a basis for reducing casing pressure risks and optimizing wellbore integrity design.

Keywords: sustained annular pressure, prefabricated micro-fracture, channeling experiment, channeling model

Introduction

The SCP in high temperature high pressure gas wells and gas storage wells is a critical technical challenge for wellbore integrity. During the production, the SCP is common, threatening wellbore integrity^[1]. When the wellbore integrity fails, the pressure in the annulus increases continuously during production, affecting safe production. In gas storage wells, cyclic injection-production significantly increases loads on tubular, raising the probability of isolation failure. Consequently, establishing a model of annular pressure is crucial for wellbore integrity optimization and SCP risk reduction. At present, theories of abnormal annular pressure formation include hydrostatic pressure loss by gelling suspension, bridge plugging, weak interface bond, and

micro-fracture/ micro-annuli. Among these, the micro-fracture / micro-annuli theory has gained attention for explaining SCP in late-stage of production. In this theory, the annular comprises three parts: the casing-cement interface (first interface), the cement sheath, and the cement-formation interface (second interface). In 2010, Che et al.^[3] developed a model to calculate annular pressure in gas well, considering effect of temperature and pressure. In 2014, Rocha et al.^[4] proposed a thermodynamic model to predict SCP. In 2019, Feng et al.^[5] derived a gas seepage rate equation in cement sheath, and they treated the liquid column segment as an elastic body, proposing an iterative correction algorithm for bulk modulus. In 2021, Du et al.^[6] introduced a multi-variable grey prediction model for real-time pressure prediction in SCP gas well. Despite many contributions have been made to simulate pressure of channeling, the law of channeling under different micro-annuli size remains unclear, and the validation methods are lacking. In this paper, a multi-scale channeling model with macroscopic channeling in micro-fracture and gas seepage in weakly bond is established, and gas flow characteristics under different micro-fracture height are investigated by experiment. The relationship between micro-fracture height and equivalent permeability is inverted. The annular pressure recovery curves are used to classify channeling paths, supporting SCP mitigation.

Micro-fracture channeling experiment

Material and device

By utilizing metal film presetting technology, micro-fracture height can be controlled. In micro-fracture, one side is covered with metal film, and the other side is cements. Micro-fracture height of cements included 10 μm , 30 μm , 50 μm , 60 μm , and 100 μm , as shown in Fig. 1. The cements is a cylindrical body with a diameter of 38mm and a length of 50mm.



Figure 1—Cements with different micro-fracture height.

The device mainly includes a control system, nitrogen cylinder, pressure sensors, gripper, gas flowmeter, data acquisition, and confining pressure tracking pump, as shown in Fig. 2.

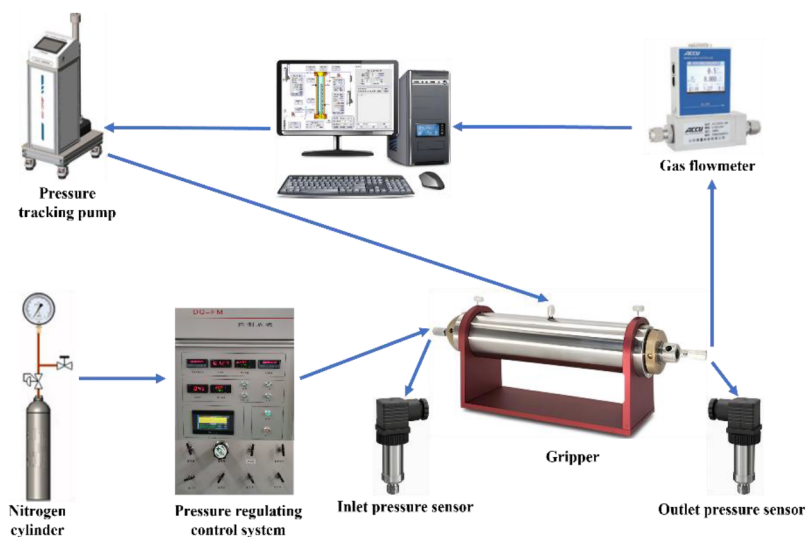


Figure 2—Channeling experiment device

Method

Preparation. Open the data collection software, and set the basic parameters. Open the nitrogen cylinder and ensure the pressure gauge meets the experimental requirements. Start the pressure tracking pump to ensure that the liquid in the pump reaches the maximum capacity.

Cements assembly. Prepare 9 cements with the same micro-fracture width. Place the cements into the gripper in accordance with numbered sequence, and tighten the gripper.

Confining pressure applying. Set the confining, and observe the outlet on the upper part of the gripper. If the liquid flows out, it indicates that the air inside the gripper has been completely discharged. Close the outlet and gradually increase the confining pressure to the set value.

Conduct experiments. Change the parameters through a pressure control system, and measure the gas flow rate through the cements. The data is collected in software.

Results

Channeling experiment under different inlet pressure and different micro-fracture height

The cements with 5 kinds of micro-fracture height (10 μm , 30 μm , 50 μm , 60 μm , 100 μm) are prepared. The experimental temperature is room temperature, and the confining pressure is 10MPa. During the experiment, the inlet pressure increases by 0.1MPa in turn. After each group of experimental pressure is stable, record the pressure and the rate. Rate versus inlet pressure is obtained for cements of different micro-fracture height, as shown in Fig. 3.

It can be seen that as the pressure difference between the two ends of the cements increases, the rate tends to increase linearly. Under the same pressure difference, the larger the micro-fracture height of the cements, the greater the rate through the cements.

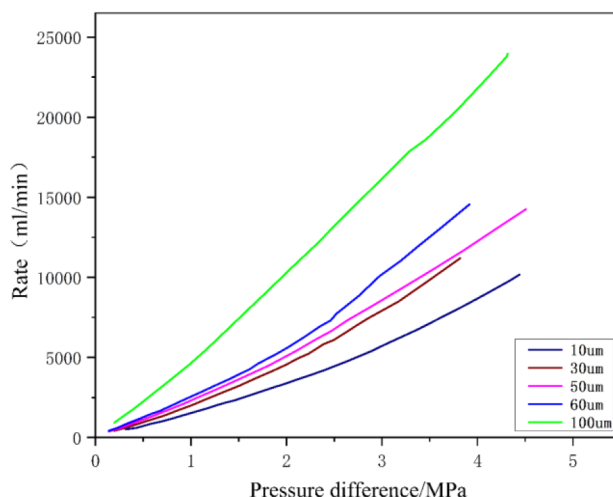


Figure 3—The relationship between pressure difference and rate of cements with different micro-fracture height.

Channeling experiment under different inlet pressure and outlet pressure

The cements with 5 kinds of micro-fracture height (10 μ m, 30 μ m, 60 μ m, 100 μ m) are prepared. Install a pressure holding valve to control the pressure of the outlet end of the cements. The experimental temperature is room temperature, and the confining pressure is 10MPa. During the experiment, the outlet pressure of the cements is controlled at 1MPa. The inlet pressure of the cements steadily increases from 1MPa to 5MPa. The rate at the outlet of cements is record. Similarly, the outlet pressure of the cements is controlled at 2MPa and 3MPa respectively, and the above experimental operations are repeated. The rate versus pressure difference under different outlet pressure are obtained, as shown in Fig. 4.

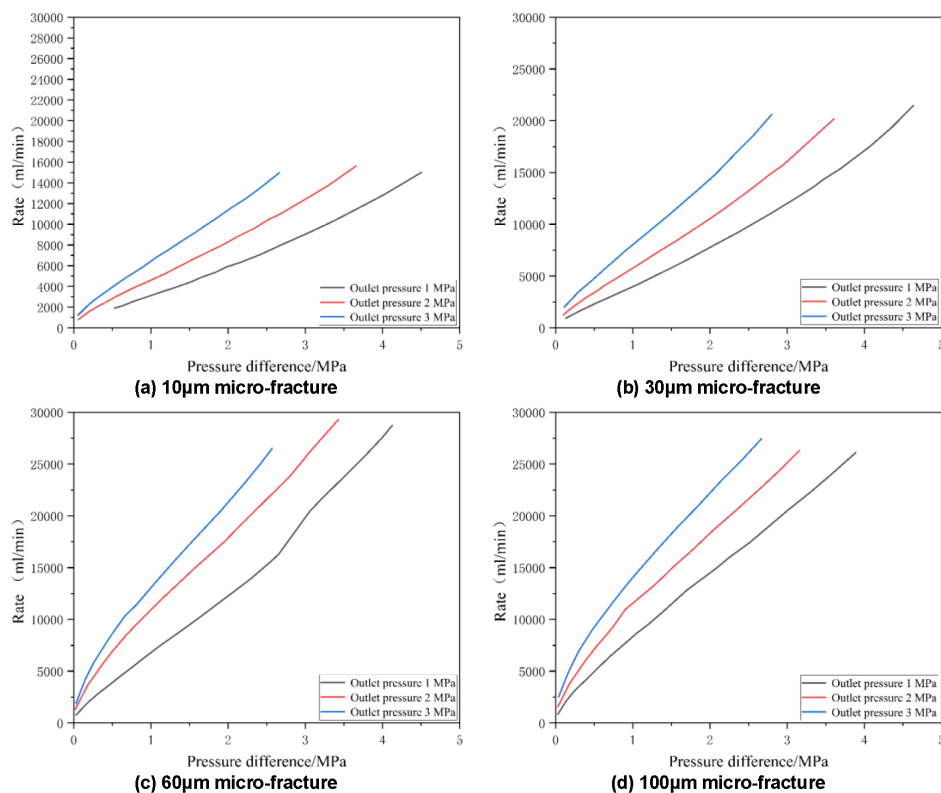


Figure 4—The gas flow rate changes with the pressure difference when the two ends of the cements with different micro-fracture height are pressurized.

When the outlet pressure of the cements remains constant, the rate gradually increases as the inlet pressure of the cements increases. When the rate is constant, the higher the outlet pressure, the smaller the pressure difference between the two ends of the cements. When the confining pressure is constant, the increase of the outlet pressure will increase the stress loaded on the cements micro-fracture, increase the deformation of the micro-fracture, and increase the rate. In this experiment, micro-fracture height of cements ranges from 10 μm to 100 μm . However, due to limited technology, the actual micro-fracture height of the cements is modified by CT. The 10 μm is corrected to 35 μm , the 30 μm is corrected to 48 μm , the 50 μm is corrected to 62 μm , and the 100 μm is corrected to 115 μm . The data fitting and theoretical calculation are based on the corrected data.

Multi-scale micro-fracture channeling model

Weakly bonded channeling path. There is a weakly bonded transition zone between the casing and the cements, and the thickness of this zone is affected by the mud cake and cement slurry properties. The permeability of the transition zone is much lower than that of the cements. Therefore, the flow resistance of gas in the weakly bonded transition zone is much lower than that in the cements.

The flow of gas through weakly bonded cements is low-speed seepage, which conforms to the linear seepage equation. During a certain distance of seepage, the temperature, gas viscosity, and compression factor do not change. The seepage can be regarded as isothermal stable Darcy seepage. The expression is:

$$v = -\frac{k}{\mu} \cdot \frac{dp}{dx} = \frac{k}{\mu} \cdot \frac{p_1^2 - p_2^2}{L} \cdot \frac{1}{2p} \quad (1)$$

Where, K is the permeability, μ is the gas viscosity, p_1 and p_2 are the pressure at both ends of the cements, L is the length of the channeling path, and P is the pressure.

To characterize the flow ability of gas through weakly bonded transition zone, K is defined as the comprehensive permeability coefficient, which combines the flow ability of gas through micro-fracture and matrix. The isothermal seepage rate of gas can be expressed as:

$$Q = \frac{43.2 K A T_{sc}}{p_{sc} T L \mu Z} (p_f^2 - p_c^2) \quad (2)$$

Where, Q is the rate at the outlet, m^3/d ; L is the total length of the cements, m ; K is the comprehensive permeability, μm^2 ; μ is the gas viscosity, $\text{mPa}\cdot\text{s}$; A is the cross-sectional area of the weakly bonded transition layer, m^2 ; T_{sc} is the standard temperature, K ; p_{sc} is the standard pressure, MPa ; T is the transient temperature of the gas, K ; p_f is the inlet pressure, MPa ; p_c is the outlet pressure, MPa .

The relationship between pressure difference and rate through cements with different micro-fracture height is applied into [equation \(2\)](#), and the comprehensive permeability of prefabricated cements can be obtained, as well as the average pressure (the average of inlet pressure and outlet pressure) of cements with different micro-fracture height, as shown in [Fig. 5](#).

It can be seen that the comprehensive permeabilities of cements with 35-115 μm micro-fracture are higher than 1 μm^2 , while the gas permeability of conventional cements is much less than 1 μm^2 . Therefore, it can be concluded that the flow behavior in the micro-fracture channeling experiment does not belong to Darcy flow.

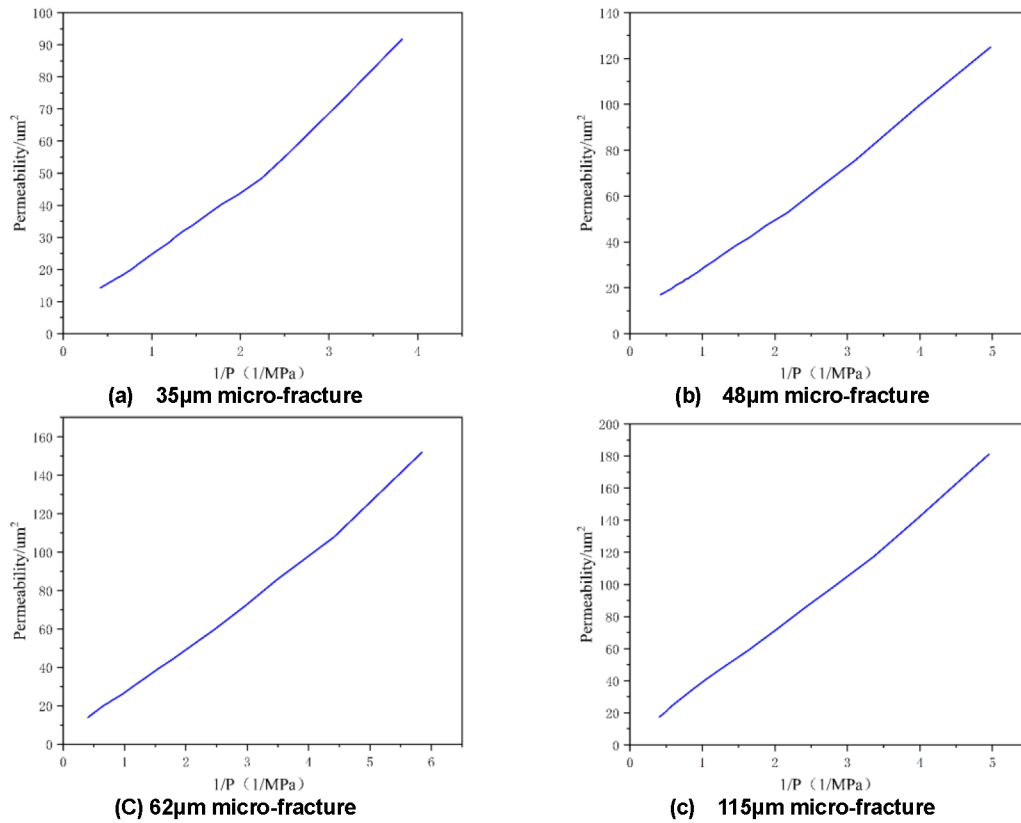


Figure 5—The comprehensive permeability of cements with different micro-fracture height under different average pressures.

Micro-fracture channeling path. The temperature or pressure changes in the wellbore cause radial displacement of the casing, especially during the pressure or temperature drop after the cement setup. The shrinkage of the cements will cause the development of micro-annuli, leading to the failure of the isolation and the formation of a channeling path.

The flow inside the micro-fracture can be regarded as the flat flow. According to the Boussinleck equation, the rate in a unit length micro-fracture can be expressed as:

$$q = \frac{h^3}{12\mu} \frac{dp}{dl} \quad (3)$$

According to the Hagen Poiseuille equation, when the micro-fracture width is , considering the influence of gravity, the rate through the micro-fracture is expressed as:

$$Q = \frac{wh^3}{12\mu L} (\Delta p - \rho g L \sin\theta) \quad (4)$$

The experimental data is applied into equation (4), and the results are shown in Fig. 6.

It can be seen that there is a positive correlation between experimental value and theoretical value, but the slope of theoretical value is inconsistent with that of experimental value. Therefore, it is necessary to modify the existing micro-fracture channeling equation to describe the flow behavior in the cements with micro-fracture.

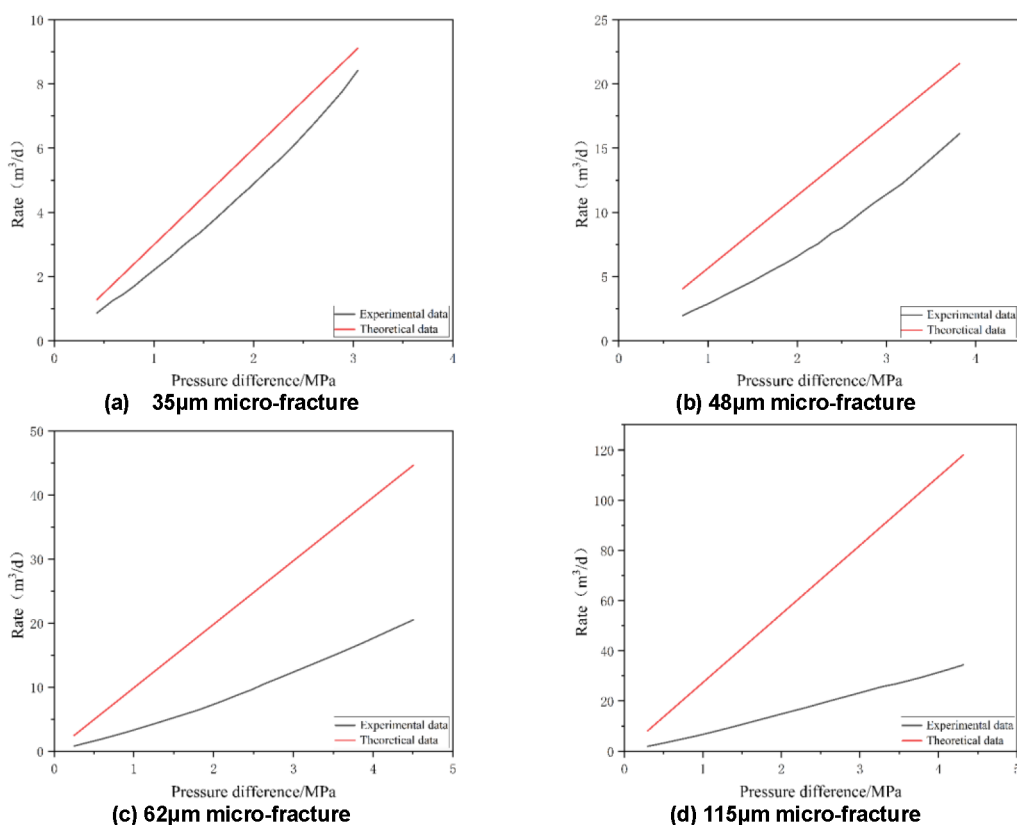


Figure 6—The theoretical value the experimental value of Hagen-Poiseuille.

Based on the comparison of theoretical and experimental values, it can be found that the pressure difference is linearly positively correlated with both theoretical and experimental values of rate. Formula fitting can be considered from the relationship between micro-fracture height and the slope angle of the two curves^[8]. The relationship between micro-fracture height and the slope angle of the two curves is shown in Fig. 7.

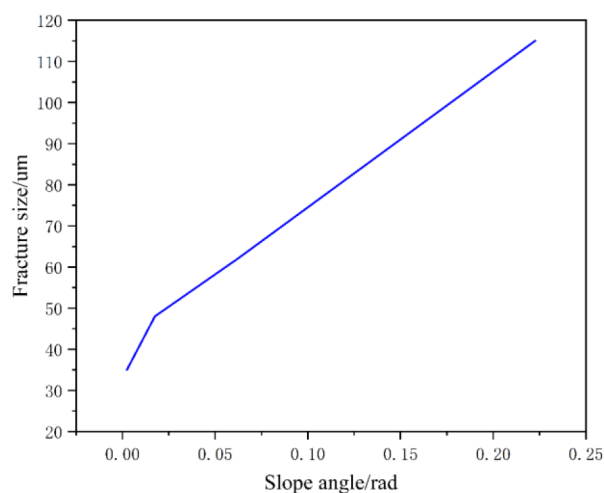


Figure 7—The relationship between micro-fracture height and the slope angle of the two curves

According to Fig. 7, the Hagen Poiseuille equation applicable to the flow behavior of cements with different micro-fracture height is obtained:

$$Q = 0.0864 \times \left(\frac{wh^3}{12\mu L} \times 10^{12} - 4.6134 \times 10^7 h^2 + 4144.6h - 0.0894 \right) \times (1000\Delta p - 0.001\rho g L \sin\theta) \quad (5)$$

Where, w is the micro-fracture width, m; h is the micro-fracture height, m; P is the gas density, kg/m³; G is the gravity, generally taken as 9.8N/kg; Δp is the difference between the inlet pressure and the outlet pressure, MPa; θ is the angle between the micro-fracture and the horizontal plane, in degrees.

The fitted theoretical values and experimental values are shown in Fig. 8. The theoretical values are in good agreement with the experimental values, indicating that the fitted Hagen Poiseuille equation can explain the flow behavior in the cements.

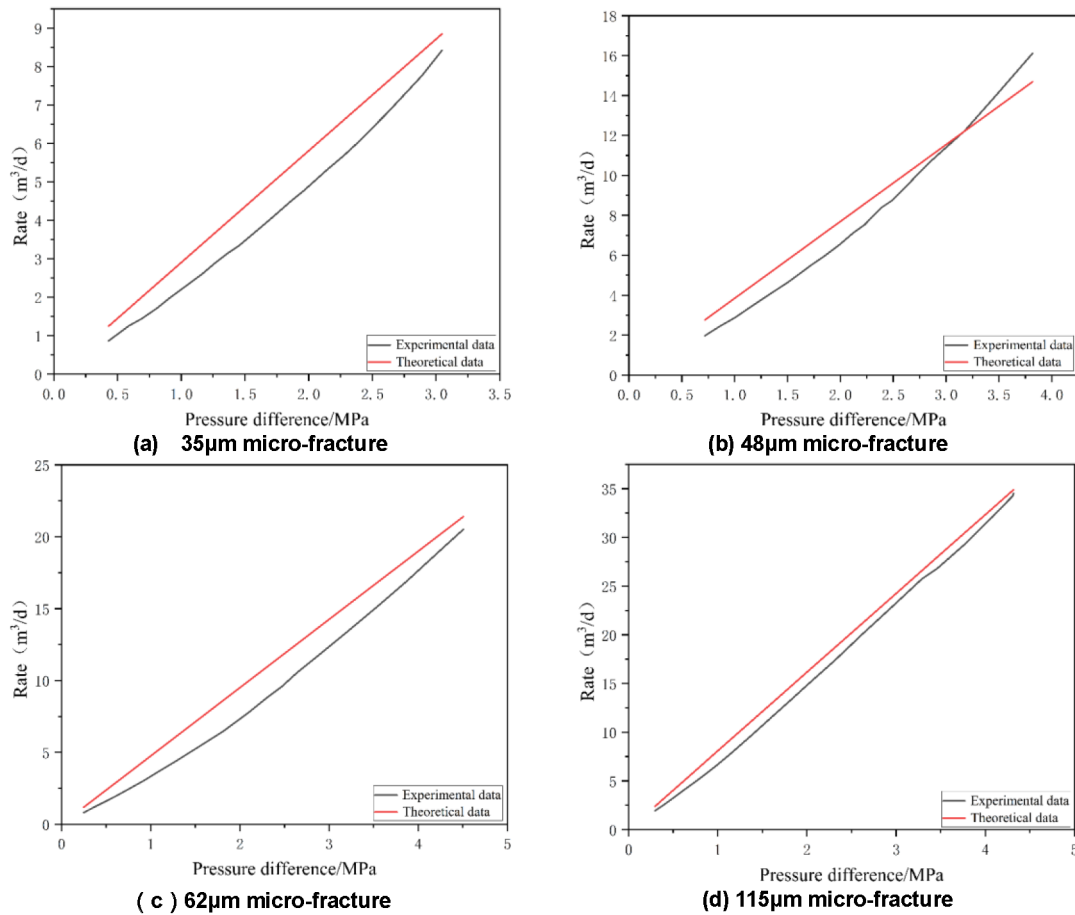


Figure 8—The fitted theoretical value and the experimental value of rate.

The multi-scale channeling model is established by integrating channeling in micro-fracture channeling path and weakly bonded channeling path.

1. Determine the micro-annuli size of each interval;
2. Determine the demarcation of the model based on the relationship between the micro-fracture height and the initial pressure of the micro-annuli;
3. Using the comprehensive permeability under various pressure to iteratively calculate the demarcation pressure;
4. Simulate the pressure of the wellhead through the model demarcation pressure.

Case Study

In the initial stage of gas production, the formation pressure of a gas storage well is 31.55MPa, and the bottom hole static temperature is 120.6 °C. The vertical depth of the well is 3075 m, and the measured depth is 3188 m, as shown in Fig. 9. The well has been operated for 3 years, and the pressure of B annulus is 6.6MPa.

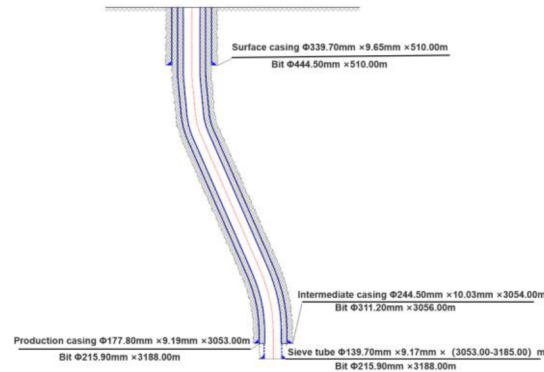


Figure 9—Well structure diagram.

The multi-scale micro-fracture channeling model is applied to analyze the wellhead pressure of gas storage wells under different micro-fracture height. The initial pressure in the micro-fracture is assumed as the formation pressure. The changes in demarcation pressure and wellhead pressure under the 10-50 μm micro-fracture height are analyzed, as shown in Fig. 10.

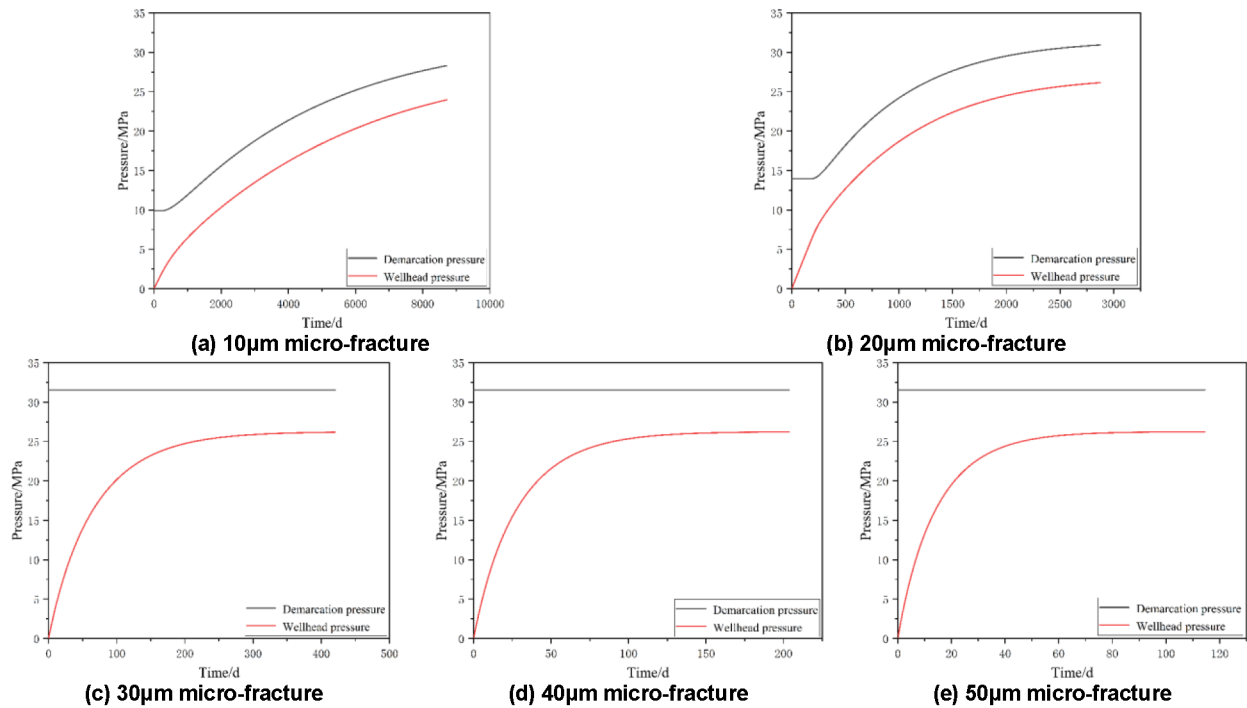


Figure 10—The relationship among the demarcation pressure, the wellhead pressure and the micro-fracture height.

It can be seen that when the micro-fracture height is less than 30 μm , the well is divided into two intervals, the lower interval is the gas seepage interval and the upper interval is the micro-fracture interval. In the seepage interval, gas firstly migrates from the bottom to the top of the interval. Due to the better permeability

of the micro-fracture interval compared to the seepage interval, gas will continue to migrate towards the wellhead. Therefore, gas will accumulate at the wellhead, and the annular pressure gradually increases. The pressure difference between the top of the seepage interval and the wellhead gradually decreases, and the growth rate of wellhead pressure decreases. When the pressure difference between the top of the seepage interval and the wellhead decreases to a certain value, gas begins to accumulate at the top of the seepage interval, and the pressure gradually increases. Finally, the pressure distribution of the wellbore tends to stabilize, and gas migration no longer occurs. When the micro-fracture height exceeds 30 μm , it can be considered that the entire cements is micro-fracture interval, and the model demarcation is the bottom.

In order to validate the model, a pressure release is carried out in the B annulus of the gas storage well. When the wellhead pressure dropped to 0, the pressure release is stop and the wellhead pressure recovery data is carried out. The wellhead pressure evaluation device is shown in Fig. 11.

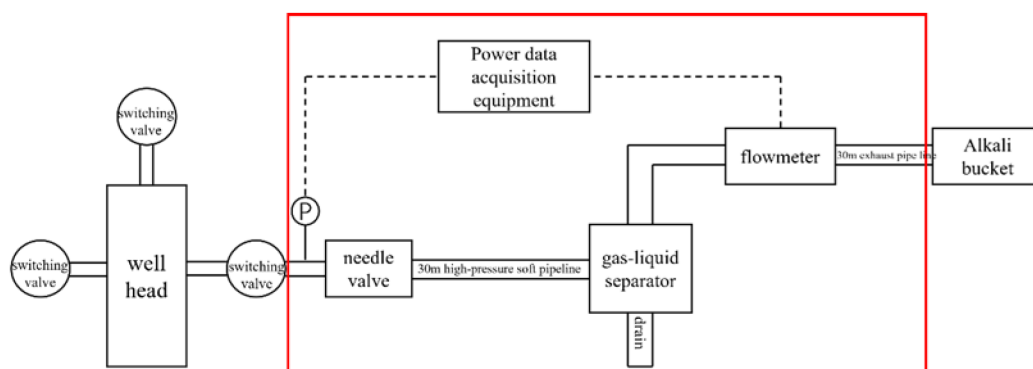


Figure 11—The wellhead pressure evaluation device.

The monitoring value and prediction value of the wellhead pressure are shown in Fig. 12. It was found that the prediction value is close to the monitoring value, indicating that the model can accurately reflect the actual recovery process of wellhead pressure in gas storage well and can be used for predicting wellhead pressure in complex oil and gas wells.

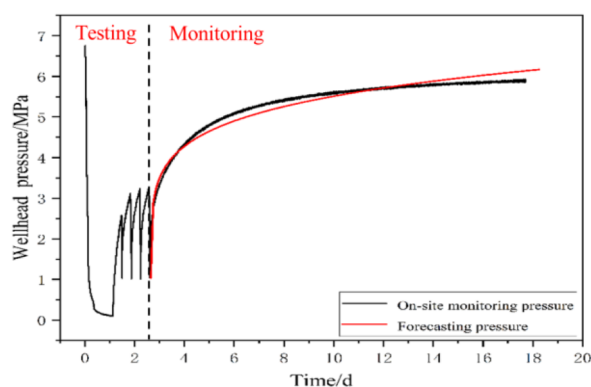


Figure 12—Wellhead pressure recovery monitoring value and model prediction value.

Conclusions

1. With the help of experimental data applied to the model, it shows that when the micro-fracture height is greater than 35 μm , the channeling does not conform to the gas seepage in weakly bonded cements.
2. The gas flow model of micro-fracture channeling path based on the Hagen Poiseuille equation is linearly positively correlated with the experimental data. The fitted theoretical values are in good

agreement with the experimental values, that indicates the model can be used for micro-fracture channeling simulation.

3. The demarcation can be determined by the micro-fracture height and pressure in multi-scale micro-fracture channeling model. The different channeling path is considered, and the wellhead pressure is determined. The simulation data is in good agreement with on-site monitoring data, indicating that the model has certain reliability and accuracy.

Acknowledgment

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